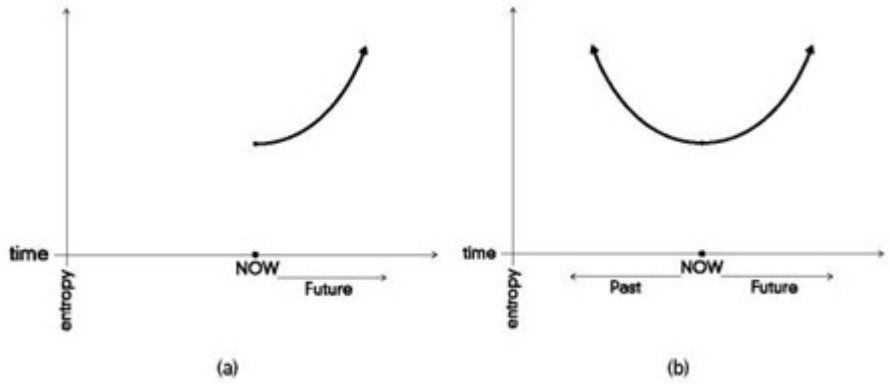


Figure 6.2 (a) As it's usually described, the second law of thermodynamics implies that entropy increases toward the future of any given moment. (b) Since the known laws of nature treat forward and backward in time identically, the second law actually implies that entropy increases both toward the future and toward the past from any given moment.

That's the essential lesson. It tells us that the entropic arrow of time is *double-headed.* From any specified moment, the arrow of entropy increase points toward the future *and* toward the past. And that makes it decidedly awkward to propose entropy as the explanation of the *one-way* arrow of experiential time.

Think about what the double-headed entropic arrow implies in concrete terms. If it's a warm day and you see partially melted ice cubes in a glass of water, you have full confidence that half an hour later the cubes will be more melted, since the more melted they are, the more entropy they have.[11] But you should have exactly the same confidence that half an hour earlier they were also more melted, since *exactly* the same statistical reasoning implies that entropy should increase toward the past. And the same conclusion applies to the countless other examples we encounter every day. Your assuredness that entropy increases toward the future— from partially dispersed gas molecules' further dispersing to partially jumbled page orders' getting more jumbled —should be matched by *exactly* the same assuredness that entropy was also higher in the past.

The troubling thing is that half of these conclusions seem to be flatout wrong. Entropic reasoning yields accurate and sensible conclusions when applied in one time direction, toward what we call the future, but